

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project, **FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform**, aims to transform traditional debt management and financial recovery processes using Artificial Intelligence and modern web technologies. The platform improves financial decision-making, settlement planning, and borrower support through intelligent automation and real-time financial analysis.

Project Overview	
Objective	The primary objective of this project is to modernize and optimize the traditional loan approval process using advanced Machine Learning techniques. The system aims to provide faster, more accurate, and data-driven loan approval decisions while minimizing manual intervention and human error.
Scope	This project focuses on analyzing, improving, and automating the loan approval workflow by integrating intelligent Machine Learning models. It covers applicant data analysis, creditworthiness prediction, risk assessment, and automated decision-making to create a more secure, scalable, and efficient lending system.
Problem Statement	
Description	Traditional loan approval systems often suffer from delays, inconsistent evaluations, manual processing errors, and inaccurate risk assessments. These inefficiencies negatively impact operational productivity, increase financial risks, and reduce customer satisfaction. Existing systems may fail to accurately evaluate applicants' repayment capabilities, resulting in poor lending decisions.

Impact	Solving these challenges will improve operational efficiency, reduce loan default risks, enhance decision accuracy, and streamline the lending process. It will also improve customer satisfaction through quicker responses, fair evaluations, and transparent loan approval decisions, contributing to organizational growth and trust.
Proposed Solution	
Approach	The proposed solution utilizes Machine Learning algorithms to analyze applicant financial data, repayment history, income patterns, credit scores, and other relevant factors to predict creditworthiness. The system continuously learns from historical loan data to improve prediction accuracy and support intelligent decision-making.
Key Features	• Machine Learning-based credit assessment model • Automated loan eligibility verification • Risk analysis and prediction • Faster loan approval process • Reduced manual errors • Real-time applicant evaluation • Secure customer data handling • User-friendly dashboard for monitoring and management • Scalable and adaptive intelligent lending platform

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i3/i5 Processor or above, Multi-core CPU
Memory	RAM specifications	Minimum 8 GB RAM
Storage	Disk space for application data, reports, logs, and database	1 TB SSD
Internet Connectivity	Required for API integration and deployment	Stable broadband internet connection

Software		
Frameworks	Frontend and Backend frameworks	React.js, FastAPI, Tailwind CSS
Libraries	Additional libraries and dependencies	SQLAlchemy, Axios, JWT, bcrypt, pandas, numpy, matplotlib, scikit-learn
AI Integration	AI and Generative AI services	Google Gemini API
Development Environment	IDE and development tools	VS Code, PyCharm, Jupyter Notebook
Database	Data storage and ORM	SQLite, PostgreSQL-ready architecture
Version Control	Source code management	Git, GitHub
Deployment Platforms	Cloud hosting services	Vercel, Render
Data		
Financial Data	User loan and EMI details	Loan amount, EMI, income, overdue duration, settlement history
AI Analysis Data	Financial health and recommendation data	AI-generated settlement suggestions and negotiation history
Data Format	Storage and processing format	CSV, JSON, SQLite Database
Data Sources	User-provided financial information and AI-generated outputs	Borrower inputs, financial analytics, Gemini AI responses